

Publication Directions

1. Purpose

This file outlines how the current project can be shaped into paper-style research outputs.

2. Potential Paper Themes

2.1 Multimodal screening paper

Possible title:

- Multimodal Learning-Disorder Screening Using Handwriting, Audio, Text, and Reading Behavior

Core contribution:

- compare several multimodal architectures on one standardized manifest
- show which modalities help most
- add explainability and educational summaries

2.2 Low-resource transfer paper

Possible title:

- Cross-Lingual Transfer for Bengali Learning-Disorder Screening in Low-Resource Educational Settings

Core contribution:

- measure English-to-Bengali transfer
- compare scratch and warm-start training
- test foundation-style adaptation

2.3 Explainable intervention paper

Possible title:

- Explainable Educational AI for Adaptive Intervention Planning in Learning-Disorder Support

Core contribution:

- connect predictions to intervention suggestions
- evaluate explainability from a teacher/parent perspective

2.4 Biomarker discovery paper

Possible title:

- Digital Biomarker Discovery for Multimodal Learning-Disorder Screening

Core contribution:

- rank candidate markers from handwriting, speech, and reading behavior
- compare feature families and interpretability

2.5 Eye-tracking paper

Possible title:

- Visual Focus and Eye-Tracking Signals for Reading Difficulty Screening

Core contribution:

- use gaze metrics and visual-focus tasks
- compare reading stability indicators

3. Suggested Paper Structure

1. Problem and motivation
1. Related work
1. System design
1. Data and annotation
1. Model families
1. Experiments
1. Results
1. Error analysis
1. Explainability and usability
1. Limitations and future work

4. Good Figures To Include

- system architecture diagram
- data flow diagram
- model family comparison table
- validation matrix with cross-validation and holdout columns
- ranking snapshot for the current best-to-worst model order

- modality-attention chart
- biomarker ranking plot
- intervention workflow diagram

5. Good Tables To Include

- manifest schema
- model catalog
- experiment matrix
- model comparison matrix
- validation matrix
- dataset summary
- per-language results
- per-modality ablation results

6. What Makes The Project Publishable

The strongest publication angles are:

- multimodal integration
- low-resource Bengali focus
- explainable educational output
- intervention planning
- local-first implementation

7. Main Publication Risks

- insufficient dataset size
- lack of external validation
- noisy labels
- unclear annotation protocol
- limited comparison baselines